University Of Jordan

Faculty of Engineering



EMBEDDED SYSTEMS

Semester: Fall 2025

Introduction:

The goal of this project is to control a servo motor using a keypad for angle input and display the angle on 7-segment displays. The user enters an angle (0–180) from the keypad, and the servo rotates accordingly.

Tools:

- Microcontroller: PIC16F877A
- 7-segment displays
- 4x3 Keypad
- Proteus Simulation
- Servo motor
- MPLAB IDE

Technical Explanation:

1. Keypad Reading:

- The 4x3 matrix keypad is connected to PORTB.
- The program scans each column by grounding it and checking the rows.
- Once a pressed key is detected, the code uses a lookup table to return the corresponding ASCII value ('0' = 0x3).
- The ASCII code is converted to a real number by subtracting '0'.

2. How the Value is Converted to an Angle:

- The user enters 3 digits (example: 1–8–0).
- These digits are converted to a full number:

angle = digit1 × 100 + digit2 × 10 + digit3

- If the user presses *, it's used as a backspace.
- If the user presses #, the value is confirmed (unless it's over 180).

3. How PWM is Calculated:

- We use Timer2 with PR2 = 155 for a 50Hz PWM (20ms period).
- The duty cycle is calculated with this formula:

  duty = (angle / 6) + 31

This value is stored in CCPR1L to control the PWM output on RC2.

ANGEL SERVO	(ANGLE/6) +31	CCPR1L VALUE
0	(0/6) +31	31
90	(90/6) +31	46
180	(180/6) +31	61

Table : Angle vs CCPR1L Value

4. 7-Segment Display:

- The angle is split into 4 digits.
- each digit is displayed using a multiplexing technique.
- a lookup table is used to display the correct segments.

5. Interrupt (RB0):

- When RB0 is pressed:

  _ The servo is toggled ON/OFF.

  _ If OFF, the PWM duty is set to 31 (0° angle).

_ If ON again, the last angle is restored.

7. Simulation Result (in Proteus):

- Entered angle appears on the 7-segment display.
- Servo rotates based on the entered angle.
- * Acts as backspace, confirms the entry.
- RB0 toggles the servo signal ON/OFF.

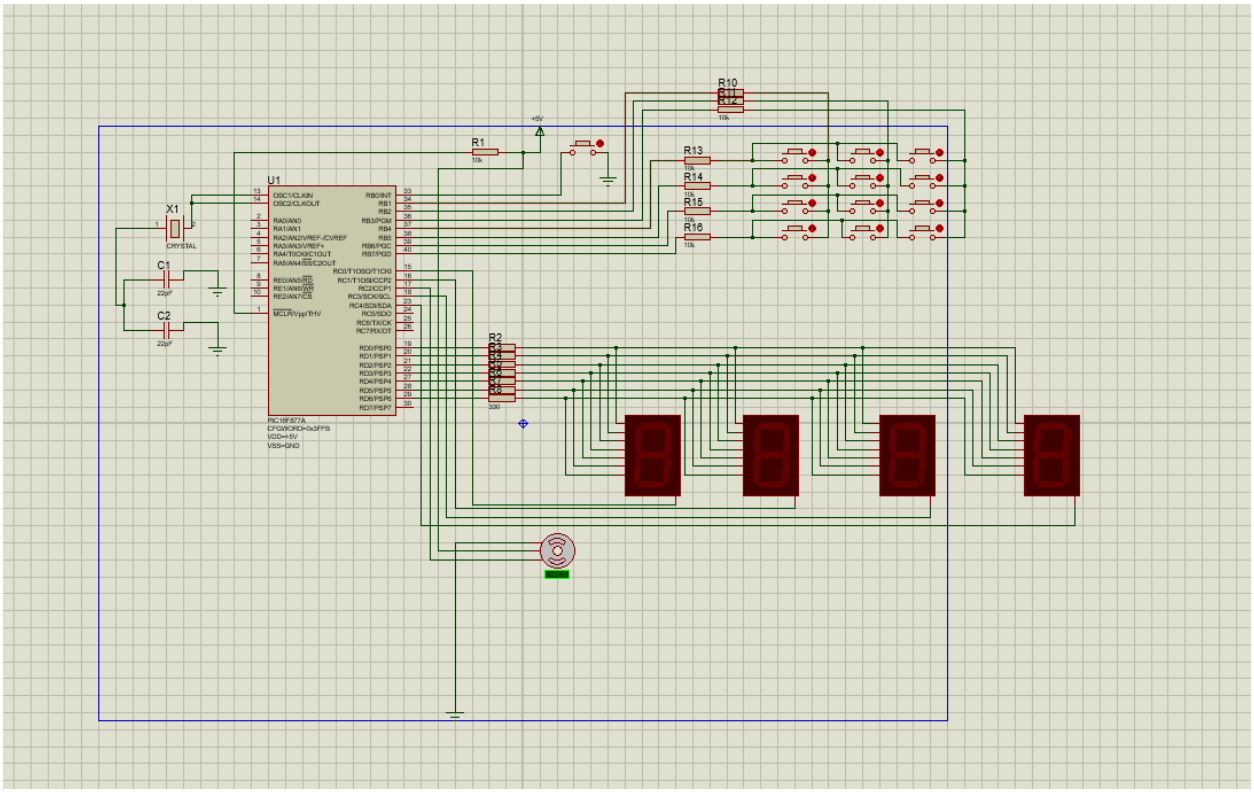


Fig: the proteus design.

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